

Implementation of Naïve Bayes Algorithm in Python

Aim

To implement the **Naïve Bayes** classification algorithm in Python and evaluate its performance using a **Confusion Matrix** and **Accuracy Score**.

Apparatus Required

- Google Colab / Jupyter Notebook
- Python 3
- Libraries:
 - NumPy
 - Pandas
 - Scikit-learn (sklearn)

Theory

Naïve Bayes is a **supervised machine learning algorithm** based on **Bayes' Theorem**. It assumes that all features are independent of each other given the class label. Despite this simple assumption, Naïve Bayes performs well for many classification problems, especially text classification and medical diagnosis.

Advantages:

- Simple and fast.
- Works well with small datasets.
- Efficient for multi-class classification.

Disadvantages:

- Assumes feature independence.
- Performance decreases if features are highly correlated.

Algorithm

1. Import the required libraries.

2. Load the Iris dataset.
3. Split the dataset into training and testing sets.
4. Train the Naïve Bayes classifier.
5. Predict the class labels for the test data.
6. Generate the confusion matrix.
7. Calculate and display the accuracy.

CODE:

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import confusion_matrix, accuracy_score
```

```
# Load Iris dataset
```

```
iris = load_iris()
```

```
X = iris.data
```

```
y = iris.target
```

```
# Split dataset
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
```

```
# Train Naive Bayes model
```

```
model = GaussianNB()
```

```
model.fit(X_train, y_train)
```

```
# Predict
```

```
y_pred = model.predict(X_test)
# Display Results
print("Predicted Classes:")
print(y_pred)
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))
print("\nAccuracy:")
print(accuracy_score(y_test, y_pred))
```

output

```
... Predicted Classes:
[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 2 2 1 1 2 0 2 0 2 2 2 2 2 0 0 0 0 1 0 0 2 1
 0 0 0 2 1 1 0 0]

Confusion Matrix:
[[19  0  0]
 [ 0 12  1]
 [ 0  0 13]]

Accuracy:
0.9777777777777777
```

Result

The Naïve Bayes algorithm was successfully implemented in Python using the Iris dataset. The classifier achieved high classification accuracy, and its performance was evaluated using the Confusion Matrix and Accuracy Score.